

Arithmetic and complexity

The least positive permanent of a sign matrix

Difficulty: challenging **Importance:** interesting to specialist**Status:** Partially resolved

Rating rationale: Challenging because divisibility alone gives no construction attaining the bound for every order; specialist importance concerns exact permanent values and discrete matrix constructions.

Independently reviewed finite cases - 2026-09-11

Complete exact certificates attain $2^{n-\lfloor \log_2(n+1) \rfloor}$ for every order $1 \leq n \leq 35$. Together with the proved universal divisibility bound, they establish the conjectured minimum at these orders, extending the cited order-20 construction range. The entire supplied matrices are checked with two coprime moduli and a proved uniqueness bound; cofactor dot products alone are not the evidence. Attainment for every order $n \geq 36$ remains unresolved by this submission.

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The [fresh certificate checks](#) complement the analytic review. The remaining universal target and its ratings are retained.

Problem statement

For $n \geq 1$, let $\Omega_n = \{-1, 1\}^{n \times n}$ and define

$$\text{per } A = \sum_{\sigma \in S_n} \prod_{i=1}^n a_{i, \sigma(i)}, \quad p_n = \min\{\text{per } A : A \in \Omega_n, \text{ per } A > 0\}.$$

The set in this minimum is nonempty, since the all-ones matrix has permanent $n!$. Is Kräuter's proposed formula

$$p_n = 2^{n-\lfloor \log_2(n+1) \rfloor}$$

valid for every n ? The problem asks for attainability of the divisibility lower bound, with no rank restriction on A .

Relevance and ratings

This is an extremal question about a basic multilinear matrix function on discrete matrices. Exact values and divisibility constraints matter for permanent computation and constructions of sign matrices. The attainability problem across all orders merits challenging difficulty; its main audience is combinatorial matrix theory.

References

- I. M. Wanless, [Permanents of matrices of signed ones](#), Linear and Multilinear Algebra **53** (2005), 427–433, §3, for attaining constructions through order twenty.
- D. Ingram and A. Razborov, *On the range of the permanent of (± 1) -matrices*, Linear Algebra Appl. **743** (2026), 271–285. The precise question is §6, Problem 3 in [arXiv:2507.09433v1](#); the [published introduction](#) gives the displayed unified formula.
- M. V. Budrevich and A. E. Guterman, *Kräuter conjecture on permanents is true*, J. Combin. Theory A **162** (2019), 306–343 ([paper](#); [journal](#)), for a different, resolved rank-dependent upper-bound conjecture.

Status check — 2026-09-10

Checked [Ingram–Razborov’s August 2026 published introduction](#), which explicitly retains the minimum-positive-value conjecture, and its [preprint §6, Problem 3](#). [Wanless \(2005\), §3, pp. 430–431](#), supplies attaining matrices for every $n \leq 20$, a proved parameter range of this target. Exact-title and minimum-positive-permanent searches found no general resolution. The [2019 Kräuter theorem](#) proves a rank-dependent upper bound; [Hunter–Kwan–Sauermann](#) proves exponential range cardinality. Neither establishes general attainment of the lower bound.